Statistical Mechanics

2009년 1학기

황 현 석

강원대학교 화학과



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Introduction to Statistical Mechanics

- Surface tension

  1) Surface tensions of liquids at 293 K (Atkins, p641)





Synoptic table 18.5* Surface tensions of liquids at 293 K

	$\gamma/(\mathrm{mN\ m^{-1}})^{\dagger}$
Benzene	28.88
Mercury	472
Methanol	22.6
Water	72.75

* More values are given in the *Data section*.
† Note that $1\ \mathrm{N\ m^{-1}} = 1\ \mathrm{J\ m^{-2}}$.

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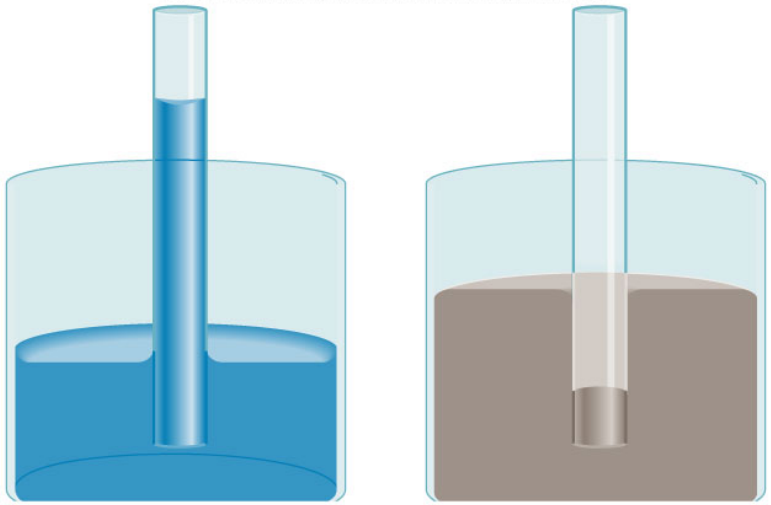
Introduction to Statistical Mechanics (*cont.*)

- Nucleation

  1) Laplace equation (Atkins, p643)

$$p_{\text{in}} = p_{\text{out}} + \frac{2\gamma}{r}$$

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(a) (b)

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Introduction to Statistical Mechanics (*cont.*)

- Nucleation

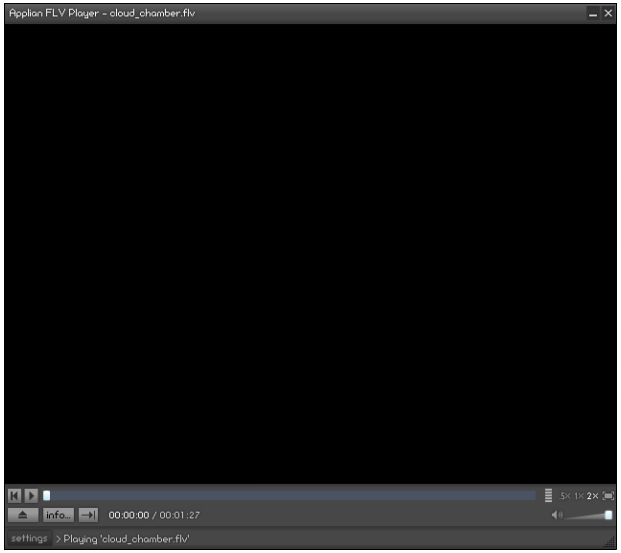
  1) Laplace equation (Atkins, p643)

$$p_{\text{in}} = p_{\text{out}} + \frac{2\gamma}{r}$$

  2) Supersaturation

  3) Cloud formation – AgI as a nucleation center

  4) Cloud chamber



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History of Statistical Mechanics

1. Caloric theory: Heat was incorrectly thought to be a conserved form of matter, a fluid called “caloric”.
   - The expansion of materials due to occupation of space by caloric.
   - The flow of heat from hot to cold objects due to repulsions between more crowded caloric particles in hot objects.
   - Temperature regarded as a direct measurement of the amount of heat.
   - Heat was considered to be conserved.
   - Heat engine by Sadi Carnot.

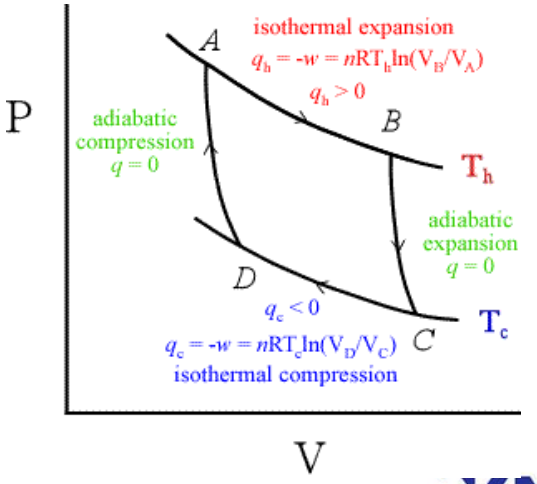
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History of Statistical Mechanics (*cont.*)

2. Sadi Carnot (1796 – 1832)

- French physicist and military engineer.
- Analyzed the efficiency of steam engines using caloric theory.
- Developed the notion of a reversible process and, in postulating that no such thing exists in nature.
- Laid the foundation for the second law of thermodynamics and initiating the science of thermodynamics.
- Carnot cycle.



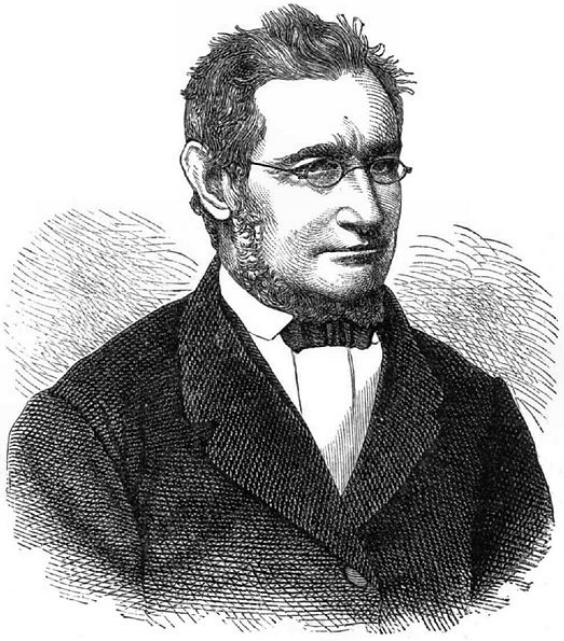


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History of Statistical Mechanics (*cont.*)

3. Julius Robert von Mayer (1814 – 1878)

- German physician.
- As an amateur scientist, wrote a paper on **the conservation of energy**, but his lack of academic training leads to its rejection.
- Speculated the mechanical equivalent of heat.
- Controversy over priority for the idea of the mechanical equivalent of heat.
- Mayer's relation: $C_p - C_v = R$



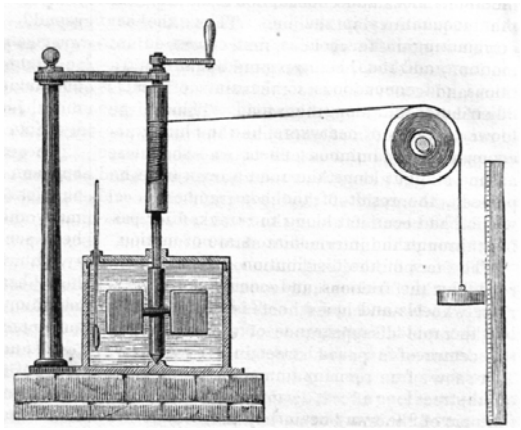
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History of Statistical Mechanics (*cont.*)

4. James Joule (1818 -1889)



- English physicist.
- Verified the mechanical equivalent of heat with experiments.



- 4.184 J = 1.0 cal.
- Speculated the “molecular theory of gases”.

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History of Statistical Mechanics (*cont.*)

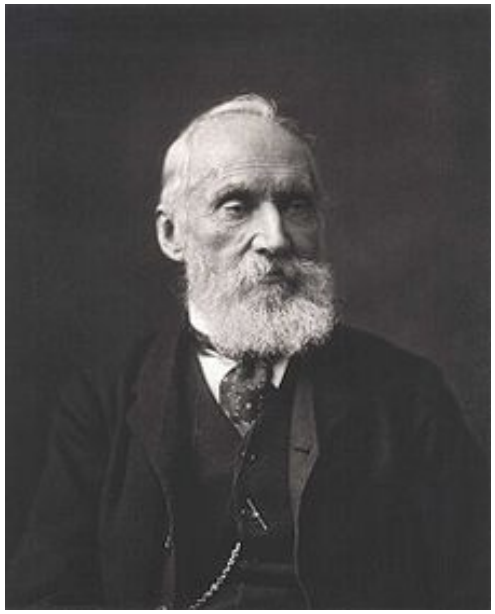
5. Hermann von Helmholtz (1821 -1894)
   - German physician and physicist.
   - Published a definitive statement of “the conservation of energy” in 1847.
   - Contributed to developing biophysics
   - Developed theories of vision and hearing.



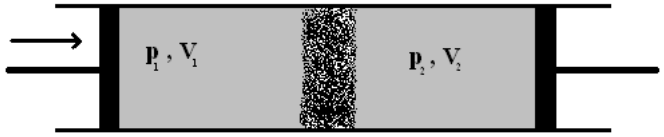
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History of Statistical Mechanics (*cont.*)

6. William Thomson “Lord Kelvin” (1824 - 1907)



- Irish-born British mathematical physicist and engineer.
- First used the term “thermodynamics” in a paper.
- Proposed and developed the absolute temperature scale.
- Collaborated with Joule in many experiments: Joule-Thomson effect

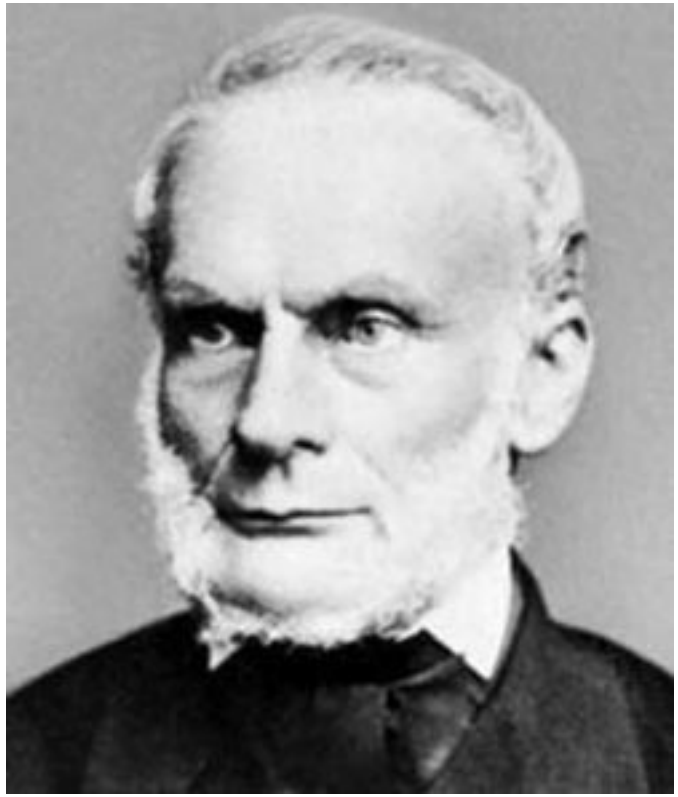


- was given the title Baron Kelvin in honor of his achievements in laying the Transatlantic cables in 1866.

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History of Statistical Mechanics (*cont.*)

7. Rudolf Clausius (1822 -1888)



- German mathematical physicist.
- Introduced the concept of entropy.
- gave the first clear joint statement of the first and second law of thermodynamics.

$$dQ = dU + PdV$$

$$dS \geq \frac{dQ}{T}$$

- Contributed to the kinetic theory of gases.

"The energy of the universe is constant."

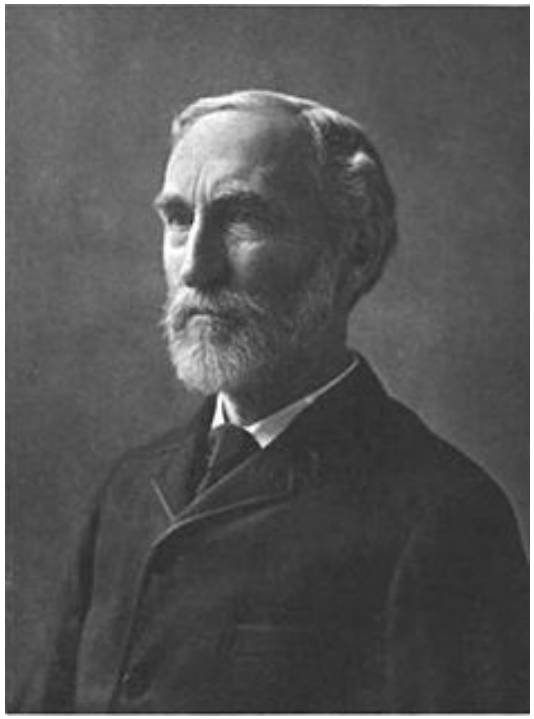
"The entropy of the universe tends to a maximum."

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History of Statistical Mechanics (*cont.*)

8. Josiah Willard Gibbs (1839 -1903)

- American theoretical physicist and chemist.
- Proposed and developed the ensemble theory and the free energy as the driving force for chemical reactions and chemical thermodynamics.
- Devised much of the theoretical foundation for chemical thermodynamics as well as physical chemistry.



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History of Statistical Mechanics (*cont.*)

9. Ludwig Boltzmann (1844 -1906)

- Austrian physicist.
- Established the connection between entropy and disorder and between entropy and probability.

$$S = k \log W$$

- Developed a molecular basis for the second law of thermodynamics and proposed the Boltzmann equation for the kinetic theory of gases.
- Founded the modern statistical Mechanics.



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